

ABDUL BASITH

Software Engineer — Full-Stack & AI Systems

MERN Stack · NestJS · PostgreSQL · RAG · LLM Integration · Multi-Tenant SaaS

Chennai, India | +91 8778685195 | abdulbasithibnhanifa@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer with hands-on full-stack and AI systems experience. Currently building AutoShipp, a multi-tenant B2B e-commerce SaaS platform, as part of a small founding engineering team, working across backend architecture, database engineering, **AI/LLM integration**, and third-party integrations. Prior to this, completed an intensive self-directed **MERN and AI systems** development period, building a full **RAG pipeline** with vector search and a secure full-stack project management platform. Comfortable working directly with production databases, live integrations, and real user-facing systems under real operational constraints.

WORK EXPERIENCE

Software Engineer — Full-Stack & AI Systems

Apr 2026 – Present · Chennai

AutoShipp — Multi-Tenant B2B E-commerce SaaS (Early-Stage Startup)

Small founding engineering team (2–3 engineers). Working on the product end-to-end: architecture, backend, database, AI integration, and third-party integrations.

System Architecture

- Architected the AutoShipp SaaS core platform: NestJS backend spanning Organizations, Identity/RBAC, Marketplace, Billing & Wallet, Communication, Shopify Integration, and Intelligence modules, built on an **80+ table PostgreSQL schema** with a Next.js frontend.
- Designed a multi-channel AI chatbot platform (WhatsApp + Instagram) with an LLM router across Gemini, Groq, Mistral, and OpenAI, and purpose-built engines for commerce, cart, recommendations, returns, persona, and safety handling.
- Designed a multi-tenant product-recommendation engine (Fit Intelligence) with A/B experimentation, analytics, Shopify catalog sync, and feature-flag infrastructure.
- Architected Verification Hub, a provider-agnostic OTP/verification platform: defined its clean-architecture workflow, tenant configuration, replay handling, and audit requirements, and produced the development plan used by the team for implementation.

Production Engineering

- Executed a **zero-downtime** PostgreSQL schema migration across 80 tables using **feature-flag-controlled dual writes**, schema parity validation, and pre-migration database snapshots — completed with **zero data loss and zero service interruption**.
- Built a transactional Shopify customer migration pipeline: paginated extraction, checkpoint/resume handling, and source-to-destination reconciliation across **7,528 source records**, with **7,525 successfully migrated** and the remaining discrepancy investigated.
- Implemented a feature-flagged JWT authentication context resolver, enabling a reversible auth cutover and incremental **RBAC migration** without breaking live traffic.
- Integrated the **WhatsApp Cloud API** in a multi-tenant setup: **HMAC-validated webhooks**, **tenant-scoped message routing**, and **Redis-based deduplication**, under a single Meta App serving multiple business accounts.
- Analyzed WhatsApp campaign performance for a live connected Shopify store (**2,900 messages sent**, **~1,500 reads**, **9 matched orders**) and calculated conversion and revenue-per-message metrics to inform campaign strategy.
- Diagnosed and fixed a **dead endpoint** and a **missing signing secret** in the **production Shopify webhook pipeline**.

Engineering Workflow

- Set up and used a **two-AI engineering workflow** for the small team: a planning AI that translated architecture decisions into structured implementation plans, and a coding AI governed by custom rule sets covering evidence-first review, git safety, and testing discipline.
- Authored **8 custom AI-agent rule files** enforcing no-assumption policy, discovery-before-implementation, and reuse-first practices to keep AI-assisted output aligned with architecture standards.

Independent Software Engineering — MERN & AI Systems

Jun 2025 – Mar 2026

Self-Directed

Self-directed engineering period focused on the MERN stack, AI systems, and full-stack architecture. Built two projects from scratch as practical output.

NeuroVault — AI Knowledge Platform

Next.js · TypeScript · Node.js · MongoDB · Supabase pgvector · Redux Toolkit · Redis · **BullMQ** · **AWS S3** | github.com/abdulbasithibnhanifa/NeuroVault

- Built a **complete RAG pipeline**: document ingestion → MiniLM vector embeddings → pgvector storage → semantic retrieval → streaming LLM response, with **multi-LLM support (Gemini, Claude, Llama)** and knowledge-graph visualization.
- Implemented an async BullMQ-based document processing pipeline for non-blocking ingestion with Redis queue management and AWS S3 storage.

DevDesk — MERN Project Management System

MongoDB · Express.js · React.js · Node.js · JWT | github.com/abdulbasithibnhanifa/DevDesk | dev-desk-app.vercel.app

- Built a full-stack project and ticket management platform with security-first auth: JWT + HttpOnly cookies + refresh-token rotation, with an Axios interceptor for automatic token refresh.
- Designed the RESTful API and MongoDB data model end to end.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript

Frontend: React.js, Next.js, Tailwind CSS, Redux Toolkit

Backend: Node.js, Express.js, NestJS, Prisma ORM, REST APIs, WebSockets, BullMQ, Webhooks

Databases: PostgreSQL, MongoDB, Supabase (pgvector), Redis, JSONB, SQL

AI Engineering: RAG, Vector Embeddings (MiniLM), Semantic Search, LLM Integration (Gemini, Claude, Llama, Groq), **Multi-LLM Routing**, Streaming

Auth & Security: JWT, HttpOnly Cookies, Refresh Token Rotation, RBAC, HMAC Validation, Feature-Flag Auth Cutover

Architecture: Multi-tenant SaaS, Clean Architecture, Event-Driven Systems, **Zero-Downtime Migrations**, **Dual-Write Strategy**

Integrations: Shopify (Customers, Products, Webhooks, OAuth, HMAC), **WhatsApp Cloud API (Meta)**, **Razorpay**

DevOps & Tools: Git, GitHub, Vercel, Render, AWS S3, GitHub Actions, Cloudflare, Sentry, Postman

EDUCATION

Bachelor of Computer Applications (BCA)

2022 – 2025

Kamaraj University College, Madurai

LANGUAGES

Tamil (Native) | English (Professional)